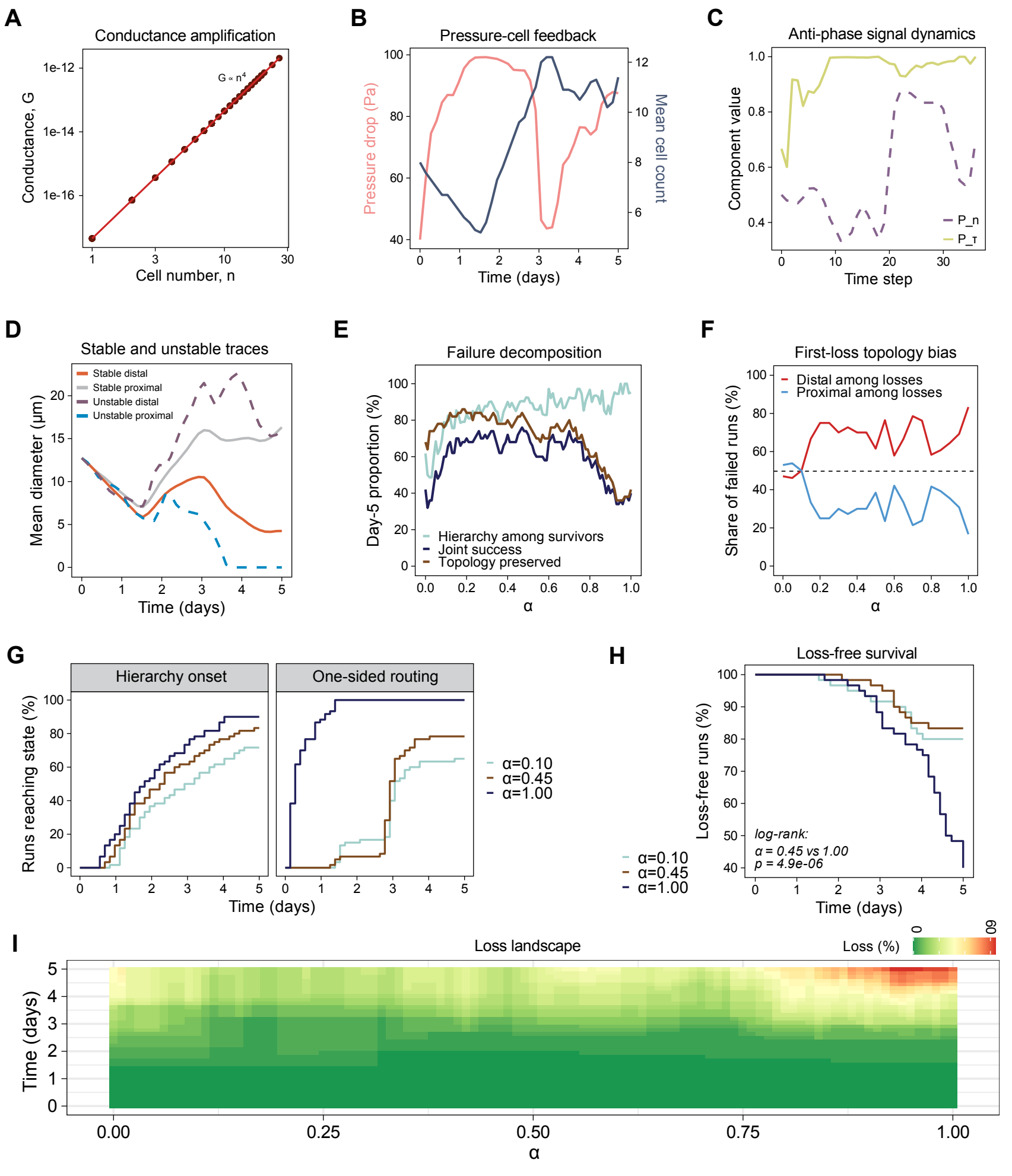
A
Conductance amplification
G ∝ n^4
Conductance, G
1e-12
1e-14
1e-16
Cell number, n
B
Pressure-cell feedback
Pressure drop (Pa)
Mean cell count
Time (days)
C
Anti-phase signal dynamics
Component value
P_n
P_τ
Time step
D
Stable and unstable traces
Stable distal
Stable proximal
Unstable distal
Unstable proximal
Mean diameter (µm)
Time (days)
E
Failure decomposition
Day-5 proportion (%)
Hierarchy among survivors
Joint success
Topology preserved
α
F
First-loss topology bias
Distal among losses
Proximal among losses
Share of failed runs (%)
α
G
Hierarchy onset
One-sided routing
Runs reaching state (%)
Time (days)
α=0.10
α=0.45
α=1.00
H
Loss-free survival
Loss-free runs (%)
log-rank:
α = 0.45 vs 1.00
p = 4.9e-06
α=0.10
α=0.45
α=1.00
Time (days)
I
Loss landscape
Loss (%)
Time (days)
α